

Customer Churn Prediction

Data Science Report — Modeling & Evaluation
Machine Learning Phase

1. Project Overview

Customer churn refers to the phenomenon where customers stop using a company's services. This report documents the machine learning phase of the Customer Churn Prediction project, covering model development, training, evaluation, and selection of the best-performing model.

Dataset Property	Value
Total Records	999,999 customers
Features (after preprocessing)	37 features
Target Variable	churn (0 = No Churn, 1 = Churn)
Class Distribution	~90% No Churn / ~10% Churn
Train / Test Split	80% / 20% (stratified)
Train Set Size	799,999 records
Test Set Size	200,000 records

2. Class Imbalance Challenge

A critical challenge in this dataset is significant class imbalance — approximately 90% of customers are labeled as No Churn, while only 10% are labeled as Churn. This means a naive model that always predicts No Churn would achieve 90% accuracy while being completely useless.

Strategies applied to handle imbalance:

Strategy	Description	Applied To
class_weight="balanced"	Penalizes misclassification of minority class more heavily	All RF, LightGBM
scale_pos_weight=9	XGBoost-specific weight for positive class	XGBoost
SMOTE	Synthetic oversampling of minority class	Tested (not selected)
F1-Score & Recall	Used as primary evaluation metrics instead of accuracy	All models

3. Models Trained

Four machine learning models were trained and evaluated on the preprocessed dataset:

Model	Description	Key Parameter
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Logistic Regression	Linear baseline model — simple and interpretable	class_weight="balanced"
Random Forest	Ensemble of 100 decision trees — majority vote	class_weight="balanced"
XGBoost	Gradient boosting — learns from previous tree errors	scale_pos_weight=9
LightGBM	Fast gradient boosting — optimized for large datasets	class_weight="balanced"

4. Evaluation Results

Models were evaluated on the 200,000-record test set using four metrics. F1-Score and Recall for the Churn class were prioritized as primary metrics due to class imbalance.

Model	F1-Score (Churn)	Recall (Churn)	Precision (Churn)	ROC-AUC
Logistic Regression	0.256	0.642	0.160	0.635
Random Forest	0.001	0.000	0.545	0.500
XGBoost	0.253	0.603	0.160	0.627
LightGBM ★	0.257	0.636	0.161	0.687

★ Selected model | Metrics calculated on test set (200,000 records)

5. SMOTE Experiment

SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data to create a balanced dataset of 720,617 samples per class. However, results showed no improvement:

Model	F1 without SMOTE	F1 with SMOTE	Verdict
Logistic Regression	0.256	0.220	Worse
XGBoost	0.253	0.020	Much Worse
LightGBM	0.257	0.010	Much Worse

Conclusion: SMOTE was not effective for this dataset. The original class_weight approach was retained.

6. Selected Model: LightGBM

LightGBM was selected as the final model based on:

- Highest ROC-AUC (0.687) — best overall discrimination between churn and no-churn
- Best F1-Score (0.257) among all models
- Optimized for large datasets — significantly faster training on 800K records
- Scalable for production deployment — suitable for the Streamlit app
- Strong gradient boosting framework — capable of learning complex patterns

Confusion Matrix Breakdown (Test Set — 200,000 records):

	Predicted: No Churn	Predicted: Churn
Actual: No Churn	114,323 (True Negative)	65,832 (False Positive)
Actual: Churn	7,219 (False Negative)	12,626 (True Positive)

The model correctly identified 12,626 churn customers (True Positives) out of 19,845 total churners — a recall of 63.6%. Only 7,219 churners were missed (False Negatives).

7. Feature Importance Analysis

LightGBM's built-in feature importance was used to identify the top drivers of customer churn:

Rank	Feature	Importance Score	Business Insight
1	monthlycharges	~270	Higher monthly bills increase churn risk
2	customer_satisfaction	~265	Low satisfaction is a strong churn signal
3	totalcharges	~200	Long-term high-spending customers may leave
4	late_payments	~200	Payment delays indicate financial dissatisfaction
5	num_service_calls	~200	Frequent support calls signal ongoing issues
6	num_services	~195	Fewer subscribed services = less engagement
7	days_since_last_interaction	~175	Inactive customers are at higher churn risk
8	contract	~170	Month-to-month contracts churn more easily
9	credit_score	~165	Lower credit scores correlate with churn
10	annual_income	~155	Income affects sensitivity to pricing

★ Top 3 features highlighted

8. Key Insights & Recommendations

Pricing is the #1 churn driver

monthlycharges and totalcharges are the top two features. The company should review pricing strategy and offer retention discounts to high-bill customers showing churn signals.

Customer satisfaction predicts churn

customer_satisfaction ranked 2nd. Proactive satisfaction surveys and follow-ups on customers scoring below average could prevent churn.

Engagement matters

days_since_last_interaction and num_services indicate that less engaged customers with fewer services are at higher risk. Cross-selling and regular communication can help.

Payment behavior is an early warning

late_payments and num_service_calls are strong indicators. A customer with repeated late payments and frequent support calls should be flagged immediately.

Month-to-month contracts are vulnerable

Customers on flexible contracts churn more easily. Incentivizing longer-term contracts with discounts could reduce churn rate significantly.

9. Conclusion

The LightGBM model was selected as the best-performing model for customer churn prediction, achieving an ROC-AUC of 0.687 and correctly identifying 63.6% of actual churners on the test set. The model has been saved and is ready for integration with the Streamlit deployment dashboard.

Metric	Value
Selected Model	LightGBM (class_weight="balanced")
ROC-AUC Score	0.687
F1-Score (Churn)	0.257
Recall (Churn)	63.6%
True Positives	12,626 churners correctly identified
Model File	models/lgbm_model.pkl